Multiple linear regression - Body Fat Index

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MHEDAS

Biomedical Statistics

Multiple linear regression - Body Fat Index

Source: https://jse.amstat.org/datasets/fat.dat.txt

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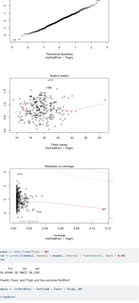
If the have a version of 3.6 or new also have to fix different body measurements of The body fat index calculated with the Brozek equation (PerfatF), the same if Index calculated with the Siri equation (PerfatS) Age (years) Weight (lbs) Height (inches) Neck circumference (cm) Chest circumference (cm) Abdomen 2 circumference (cm) Hip circumference (cm) Thigh circumference (cm) Knee circumference (cm) Ankle circumference (cm) Biceps (extended, relaxed) circumference (cm) Forearm circumference (cm) Wrist circumference (cm) The first column is the density, the fat-free weight, the fat mass, the percentage of fat, the density, the fat-free weight, the fat mass, the percentage of fat, the density, the fat-free weight, the fat mass, the percentage of fat. The last column is the density, the fat-free weight, the fat mass, the percentage of fat. The density is independent to the body fat index using simple technical measurements.

```
# Libraries
# ------------------------------------------------------------
library(car)      # Residual plots

Loading required package: carData

library(dplyr)    # Tidverse

Attaching package: 'dplyr'

The following object is masked from 'package:car':

    recode

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union

library(ggpubr)   # Tidverse
library(tidyverse)# Tidverse
library(lmerTest) # Mixed effects

Welcome to lmerTest!
Type 'step(model)' for important information if you filter this package's results.
See 'ls("package:lmerTest")' for
help.

library(effects)  # Plot effects

Lattice Cheat Sheet by effectsTheme!
Type 'trellis.par.set(theme=effectsTheme)'

library(caret)    # CARET
library(caret)    # Cross validation

Loading required package: lattice

library(mice)     # multiple imputation

Attaching package: 'mice'

The following object is masked from 'package:stats':

    filter

The following objects are masked from 'package:base':

    cbind, rbind

library(MasterMissing) # Generate missing values

# Data
# ------------------------------------------------------------
df <- read.table(file=paste("Desktop/fat.dat",sep=""),sep="",col.names=c("density","PERFAT","Age(yrs)","Weight","Height","Neck","Chest","Abdomen","Hip","Thigh","Knee","Ankle","Biceps (extd)","Biceps (flex)","Forearm","Wrist"))

head(df)

# A tibble: 6 x 16
  density PERFAT `Age(yrs)` Weight Height Neck Chest Abdomen  Hip Thigh Knee
    <dbl>  <dbl>     <dbl>  <dbl>  <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl>
1  1.0708   12.3        23  154.1   67.7  36.2  93.1   85.2  94.5  59.0  37.3
2  1.0853    6.1        22  173.1   72.0  38.5  93.6   83.0  98.7  58.7  37.3
3  1.0414   25.3        22  154.1   66.0  34.0  95.1   87.9  99.4  59.7  38.1
4  1.0267   29.9        26  148.7   64.5  34.4  95.5   90.9  99.1  59.4  38.9
5  1.0494   19.1        24  177.1   72.0  37.4  101.8  94.4  99.3  62.9  38.3
6  1.0364   26.1        24  148.7   69.5  34.4  97.1   90.7  99.3  58.3  37.3
# ... with 5 more variables: Ankle <dbl>, Biceps (extd) <dbl>, Forearm <dbl>,
# Wrist <dbl>, Biceps (flex) <dbl>

Important for you PerFatF (percentage of body fat)

summary(df)

     density          PERFAT        Age(yrs)        Weight          Height
 Min.   :0.995   Min.   : 0.0   Min.   :22.0   Min.   :118.5   Min.   :29.5
 1st Qu.:1.041   1st Qu.:15.4   1st Qu.:35.0   1st Qu.:147.0   1st Qu.:68.3
 Median :1072.8  Median :19.00  Median :43.00  Median :154.0   Median :70.0
 Mean   :1.055   Mean   :18.94  Mean   :44.88  Mean   :154.3   Mean   :70.3
 3rd Qu.:1069.5  3rd Qu.:24.10  3rd Qu.:54.0   3rd Qu.:161.5   3rd Qu.:72.5
 Max.   :1.109   Max.   :45.1   Max.   :81.0   Max.   :363.1   Max.   :77.5
     Neck           Chest          Abdomen          Hip            Thigh
 Min.   :31.1   Min.   :79.3   Min.   :69.4   Min.   :85.0   Min.   :47.2
 1st Qu.:36.4   1st Qu.:94.5   1st Qu.:88.3   1st Qu.:95.5   1st Qu.:58.1
 Median :38.00  Median :99.30  Median :90.70  Median :99.30  Median :59.00
 Mean   :37.99  Mean   :100.8  Mean   :92.56  Mean   :99.90  Mean   :59.41
 3rd Qu.:39.90  3rd Qu.:107.2  3rd Qu.:101.5  3rd Qu.:103.5  3rd Qu.:62.5
 Max.   :51.2   Max.   :136.2  Max.   :148.1  Max.   :147.7  Max.   :87.3
     Knee           Ankle          Biceps (extd)   Biceps (flex)    Forearm
 Min.   :33.5   Min.   :19.1   Min.   :24.8   Min.   :26.2   Min.   :21.0
 1st Qu.:36.5   1st Qu.:20.9   1st Qu.:30.3   1st Qu.:28.7   1st Qu.:27.3
 Median :38.50  Median :22.00  Median :32.00  Median :30.00  Median :28.7
 Mean   :38.59  Mean   :22.16  Mean   :32.27  Mean   :30.41  Mean   :28.54
 3rd Qu.:39.8   3rd Qu.:23.3   3rd Qu.:34.3   3rd Qu.:32.3   3rd Qu.:30.3
 Max.   :49.1   Max.   :33.9   Max.   :45.0   Max.   :45.1   Max.   :34.9
     Wrist
 Min.   :15.8
 1st Qu.:17.7
 Median :18.30
 Mean   :18.229
 3rd Qu.:18.8
 Max.   :21.9

## Correlation bivariate analysis

pairs(df)
```

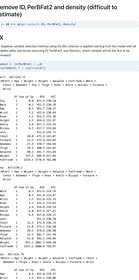


- PerFatF and Z are highly correlated
- Weight and circumference with chest and abdomen
- Height outlier
- weight outlier

EX

Let's create the file tested and available on the virtual campus. Apply the function residualPlots of the package car to the model with the three predictors PerFatF, Chest, and Thigh and the square root of PerFatF. With a significance level of 5%, conclude which predictors need a quadratic transformation.

```
model <- lm(PerFatF1 ~ PerFatF2 + Chest + Thigh, data=df)
residualPlots(model)

                 Tukey test Pr(>|Tukey test|)
PerFatF2         -2.0000     0.0455*
Chest            -2.8000     0.0051**
Thigh            -1.9000     0.0574+
Tukey test       -3.0000     0.0027**
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.025 on 240 degrees of freedom
  (1 observation deleted due to missing relevant variable(s))
F-statistic: 14.4 on 3 and 240 DF,  p-value: < 2.2e-16

plot(model)
```




```
model <- lm(PerFatF1 ~ Thigh, df)
pred <- predict(model, newdata = model, interval = "confidence", level = 0.95)
pred

       fit     lwr     upr
1 15.6144 15.5822 15.6467

PerFatF1 Chest, and Thigh and the outcome PerFatF2

model <- lm(PerFatF2 ~ PerFatF1 + Chest + Thigh, df)

plot(model)

PerFatF1 Chest  Thigh
1 18.6144 0.23000 0.27000

model <- lm(PerFatF2 ~ Thigh, data = df)
newdata <- data.frame(Thigh = 60)
pred <- predict(model,
                newdata = newdata,
                interval = "confidence",
                level = 0.95)
pred

       fit     lwr     upr
1 15.6144 15.5822 15.6467

## EX Chest data

## Remove ID, PerFatF2 and density (difficult to estimate)

df <- df[-c(1,2,3)]
```

EX

If a dependent variable selection method using the BIC criterion is applied starting from the model with all predictors (after previously removing ID, PerFatF2, and Density), which variable will be the first to be eliminated?

```
model <- lm(PerFatF1 ~ .)
step(model, k = log(nrow(df)))

Start:  AIC=-2261.74
PerFatF1 ~ Age + Weight + Height + Neck + Chest + Abdomen +
    Hip + Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq    RSS    AIC
- Hip    1   0.0 [illegible]  [illegible]
- Age    1   0.0 [illegible]  [illegible]
- Weight 1   0.1 [illegible]  [illegible]
- Knee   1   0.1 [illegible]  [illegible]
- Height 1   0.1 [illegible]  [illegible]
- Biceps 1   0.3 [illegible]  [illegible]
- Chest  1   0.4 [illegible]  [illegible]
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]


```
model <- data.frame(Thigh = 60)
pred <- predict(model, newdata = model, interval = "confidence", level = 0.95)
pred

       fit   lwr   upr
1 15.6144 15.5822 15.6467

PerFatF1 Chest, and Thigh and the outcome PerFatF2

model <- lm(PerFatF2 ~ PerFatF1 + Chest + Thigh, df)

plot(model)

PerFatF1 Chest Thigh
1 18.6144 0.23000 0.27000

model <- lm(PerFatF2 ~ Thigh, data = df)
newdata <- data.frame(Thigh = 60)
pred <- predict(model,
                newdata = newdata,
                interval = "confidence",
                level = 0.95)
pred

       fit   lwr   upr
1 15.6144 15.5822 15.6467

Chest data

Remove ID, PerFatF2 and density (difficult to estimate)

df <- df[-c(1,2,3)]

EX

If a dependent variable selection method using the BIC criterion is applied starting from the model with all predictors (after previously removing ID, PerFatF2, and Density), which variable will be the first to be eliminated?

model <- lm(PerFatF1 ~ .)
step(model, k = log(nrow(df)))

Start: AIC=-2261.74
PerFatF1 ~ Age + Weight + Height + Neck + Chest + Abdomen +
    Hip + Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Hip   1   0.0 [illegible] [illegible]
- Age   1   0.0 [illegible] [illegible]
- Weight 1   0.1 [illegible] [illegible]
- Knee   1   0.1 [illegible] [illegible]
- Height 1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- Hip   1   [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Age + Weight + Height + Neck + Chest + Abdomen +
    Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Age   1   0.0 [illegible] [illegible]
- Weight 1   0.1 [illegible] [illegible]
- Knee   1   0.1 [illegible] [illegible]
- Height 1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Weight + Height + Neck + Chest + Abdomen +
    Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Weight 1   0.1 [illegible] [illegible]
- Knee   1   0.1 [illegible] [illegible]
- Height 1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Height + Neck + Chest + Abdomen + Thigh +
    Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Height 1   0.1 [illegible] [illegible]
- Knee   1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Knee   1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Ankle + Wrist + Forearm

        Df Sum of Sq   RSS   AIC
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Abdomen + Thigh + Ankle +
    Wrist + Forearm

        Df Sum of Sq   RSS   AIC
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Thigh + Ankle + Wrist +
    Forearm

        Df Sum of Sq   RSS   AIC
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Ankle + Wrist + Forearm

        Df Sum of Sq   RSS   AIC
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Wrist + Forearm

        Df Sum of Sq   RSS   AIC
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Forearm

        Df Sum of Sq   RSS   AIC
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen


```
model <- data.frame(Thigh = 60)
pred <- predict(model, newdata = model, interval = "confidence", level = 0.95)
pred

       fit     lwr     upr
1 15.6144 15.5822 15.6467

PerFatF1 Chest, and Thigh and the outcome PerFatF2

model <- lm(PerFatF2 ~ PerFatF1 + Chest + Thigh, df)

plot(model)

PerFatF1 Chest  Thigh
1 18.6144 0.23000 0.27000

model <- lm(PerFatF2 ~ Thigh, data = df)
newdata <- data.frame(Thigh = 60)
pred <- predict(model,
                newdata = newdata,
                interval = "confidence",
                level = 0.95)
pred

       fit     lwr     upr
1 15.6144 15.5822 15.6467

## Chest data

## Remove ID, PerFatF2 and density (difficult to estimate)

df <- df[-c(1,2,3)]

## EX

If a dependent variable selection method using the BIC criterion is applied starting from the model with all predictors (after previously removing ID, PerFatF2, and Density), which variable will be the first to be eliminated?

model <- lm(PerFatF1 ~ .)
step(model, k = log(nrow(df)))

Start:  AIC=[illegible]
PerFatF1 ~ Age + Weight + Height + Neck + Chest + Abdomen +
    Hip + Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq    RSS    AIC
- Hip    1   0.0 [illegible]  [illegible]
- Age    1   0.0 [illegible]  [illegible]
- Weight 1   0.1 [illegible]  [illegible]
- Knee   1   0.1 [illegible]  [illegible]
- Height 1   0.1 [illegible]  [illegible]
- Biceps 1   0.3 [illegible]  [illegible]
- Chest  1   0.4 [illegible]  [illegible]
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Age + Weight + Height + Neck + Chest + Abdomen +
    Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq    RSS    AIC
- Age    1   0.0 [illegible]  [illegible]
- Weight 1   0.1 [illegible]  [illegible]
- Knee   1   0.1 [illegible]  [illegible]
- Height 1   0.1 [illegible]  [illegible]
- Biceps 1   0.3 [illegible]  [illegible]
- Chest  1   0.4 [illegible]  [illegible]
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Weight + Height + Neck + Chest + Abdomen +
    Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq    RSS    AIC
- Weight 1   0.1 [illegible]  [illegible]
- Knee   1   0.1 [illegible]  [illegible]
- Height 1   0.1 [illegible]  [illegible]
- Biceps 1   0.3 [illegible]  [illegible]
- Chest  1   0.4 [illegible]  [illegible]
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Height + Neck + Chest + Abdomen + Thigh +
    Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq    RSS    AIC
- Height 1   0.1 [illegible]  [illegible]
- Knee   1   0.1 [illegible]  [illegible]
- Biceps 1   0.3 [illegible]  [illegible]
- Chest  1   0.4 [illegible]  [illegible]
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq    RSS    AIC
- Knee   1   0.1 [illegible]  [illegible]
- Biceps 1   0.3 [illegible]  [illegible]
- Chest  1   0.4 [illegible]  [illegible]
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq    RSS    AIC
- Biceps 1   0.3 [illegible]  [illegible]
- Chest  1   0.4 [illegible]  [illegible]
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Ankle + Wrist + Forearm

        Df Sum of Sq    RSS    AIC
- Chest  1   0.4 [illegible]  [illegible]
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Abdomen + Thigh + Ankle +
    Wrist + Forearm

        Df Sum of Sq    RSS    AIC
- Neck   1   0.4 [illegible]  [illegible]
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Thigh + Ankle + Wrist +
    Forearm

        Df Sum of Sq    RSS    AIC
- Thigh  1   0.4 [illegible]  [illegible]
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Ankle + Wrist + Forearm

        Df Sum of Sq    RSS    AIC
- Ankle  1   0.4 [illegible]  [illegible]
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Wrist + Forearm

        Df Sum of Sq    RSS    AIC
- Wrist  1   0.6 [illegible]  [illegible]
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Forearm

        Df Sum of Sq    RSS    AIC
- Forearm 1  0.8 [illegible]  [illegible]
- Abdomen 1  [illegible]  [illegible]  [illegible]
- PerFatF1 1 [illegible]  [illegible]  [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen


```
model <- data.frame(Thigh = 60)
pred <- predict(model, newdata = model, interval = "confidence", level = 0.95)
pred

       fit   lwr   upr
1 15.6144 15.5822 15.6467

PerFatF1 Chest, and Thigh and the outcome PerFatF2

model <- lm(PerFatF2 ~ PerFatF1 + Chest + Thigh, df)

plot(model)

PerFatF1 Chest Thigh
1 18.6144 0.23000 0.27000

model <- lm(PerFatF2 ~ Thigh, data = df)
newdata <- data.frame(Thigh = 60)
pred <- predict(model,
                newdata = newdata,
                interval = "confidence",
                level = 0.95)
pred

       fit   lwr   upr
1 15.6144 15.5822 15.6467

Chest data

Remove ID, PerFatF2 and density (difficult to estimate)

df <- df[-c(1,2,3)]

EX

If a dependent variable selection method using the BIC criterion is applied starting from the model with all predictors (after previously removing ID, PerFatF2, and Density), which variable will be the first to be eliminated?

model <- lm(PerFatF1 ~ .)
step(model, k = log(nrow(df)))

Start: AIC=[illegible]
PerFatF1 ~ Age + Weight + Height + Neck + Chest + Abdomen +
    Hip + Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Hip   1   0.0 [illegible] [illegible]
- Age   1   0.0 [illegible] [illegible]
- Weight 1   0.1 [illegible] [illegible]
- Knee   1   0.1 [illegible] [illegible]
- Height 1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Age + Weight + Height + Neck + Chest + Abdomen +
    Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Age   1   0.0 [illegible] [illegible]
- Weight 1   0.1 [illegible] [illegible]
- Knee   1   0.1 [illegible] [illegible]
- Height 1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Weight + Height + Neck + Chest + Abdomen +
    Thigh + Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Weight 1   0.1 [illegible] [illegible]
- Knee   1   0.1 [illegible] [illegible]
- Height 1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Height + Neck + Chest + Abdomen + Thigh +
    Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Height 1   0.1 [illegible] [illegible]
- Knee   1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Knee + Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Knee   1   0.1 [illegible] [illegible]
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Ankle + Biceps + Forearm +
    Wrist

        Df Sum of Sq   RSS   AIC
- Biceps 1   0.3 [illegible] [illegible]
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Chest + Abdomen + Thigh +
    Ankle + Wrist + Forearm

        Df Sum of Sq   RSS   AIC
- Chest 1   0.4 [illegible] [illegible]
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Neck + Abdomen + Thigh + Ankle +
    Wrist + Forearm

        Df Sum of Sq   RSS   AIC
- Neck   1   0.4 [illegible] [illegible]
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible] [illegible] [illegible]
- PerFatF1 1 [illegible] [illegible] [illegible]

Step: AIC=[illegible]
PerFatF1 ~ Abdomen + Thigh + Ankle + Wrist +
    Forearm

        Df Sum of Sq   RSS   AIC
- Thigh 1   0.4 [illegible] [illegible]
- Ankle 1   0.4 [illegible] [illegible]
- Wrist 1   0.6 [illegible] [illegible]
- Forearm 1 0.8 [illegible] [illegible]
- Abdomen 1 [illegible]

```
Rows:  1,338
Columns:  20
$ PerBfatF1   <dbl>
$ Fat         <dbl>
$ Sex         <fct>
$ Age         <dbl>
$ Height      <dbl>
$ Weight      <dbl>
$ Diabetes    <fct>
$ Triglycerides <dbl>
$ Cholesterol <dbl>
$ Hdl         <dbl>
$ Ldl         <dbl>
$ Chol_Hdl    <dbl>
$ Ldl_Hdl     <dbl>
$ Smoking     <fct>
$ Drinking    <fct>
$ Fasting     <fct>
$ Hypertension <fct>
$ Stroke      <fct>
$ Cancer      <fct>
$ Bmi         <dbl>
```

no outlier

```
summarize(df)
```

```
#>  PerBfatF1     Fat        Sex          Age          Height
#>  Min.   : 5.40   Min.   :0.000   Male   :563   Min.   :135.0
#>  1st Qu.:21.00   1st Qu.:0.000   Female :775   1st Qu.:156.0
#>  Median :28.00   Median :1.000                 Median :163.0
#>  Mean   :28.04   Mean   :0.561                 Mean   :163.1
#>  3rd Qu.:35.00   3rd Qu.:1.000                 3rd Qu.:170.0
#>  Max.   :61.00   Max.   :1.000                 Max.   :195.0
#>                                                NA's   :  1
#>     Weight     Diabetes   Triglycerides   Cholesterol       Hdl
#>  Min.   :35.0   No  :1246   Min.   :  36.0   Min.   :  75.0   Min.   :16.0
#>  1st Qu.:65.0   Yes :  92   1st Qu.:  79.0   1st Qu.: 167.0   1st Qu.:48.0
#>  Median :76.0                Median : 106.0   Median : 199.0   Median :57.0
#>  Mean   :76.9                Mean   : 122.1   Mean   : 199.7   Mean   :56.9
#>  3rd Qu.:88.0                3rd Qu.: 151.0   3rd Qu.: 232.0   3rd Qu.:67.0
#>  Max.   :148.0               Max.   :1196.0   Max.   :413.0   Max.   :116.0
#>                              NA's   :  14     NA's   :  14     NA's   :  14
#>      Ldl     Chol_Hdl      Ldl_Hdl     Smoking    Drinking   Fasting
#>  Min.   : 46.0   Min.   : 1.00   Min.   : 0.50   No  :1117   No  :1117
#>  1st Qu.:114.0   1st Qu.: 3.00   1st Qu.: 2.00   Yes :  221   Yes :  221
#>  Median :141.0   Median : 4.00   Median : 2.60
#>  Mean   :140.8   Mean   : 4.24   Mean   : 2.79
#>  3rd Qu.:167.0   3rd Qu.: 5.00   3rd Qu.: 3.40
#>  Max.   :413.0   Max.   :15.00   Max.   :15.00
#>  NA's   :  14    NA's   :  14    NA's   :  14
#>  Hypertension Stroke  Cancer      Bmi
#>  No  :1117    No  :1324   No  :1287   Min.   : 11.40
#>  Yes :  221    Yes :  14   Yes :  51   1st Qu.: 23.60
#>                                        Median : 27.40
#>                                        Mean   : 27.69
#>                                        3rd Qu.: 31.40
#>                                        Max.   : 71.40
#>                                        NA's   :  14
```

Model fit

```
#================================================================================
```

```
mod0 <- lm(PerBfatF1 ~ 1, df = df)
summary(mod0)
```

```
Call:
lm(formula = PerBfatF1 ~ 1, data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-22.540  -6.540   0.150   7.150  33.150

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   28.040      0.193  145.3   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.064 on 1324 degrees of freedom
Multiple R-squared,  0.0e+00,	Adjusted R-squared:  0.0e+00
F-statistic: NaN on 0 and 1324 DF,  p-value: 1
```

Automatic variable selection

```
#================================================================================
```

```
mod0 <- stepAIC(mod0, k = log(nrow(df)))# AIC criterion because of explanatory goal
```

```
Start:  AIC=6997.7
PerBfatF1 ~ 1

           Df Sum of Sq    RSS    AIC
<none>                 66.7 6997.7
+ Age       1     1.12 65.6 7000.8
+ Sex       1     1.12 65.6 7000.8
+ Height    1     1.12 65.6 7000.8
+ Weight    1     1.12 65.6 7000.8
+ Diabetes  1     1.12 65.6 7000.8
+ Triglycerides 1     1.12 65.6 7000.8
+ Cholesterol 1     1.12 65.6 7000.8
+ Hdl       1     1.12 65.6 7000.8
+ Ldl       1     1.12 65.6 7000.8
+ Chol_Hdl  1     1.12 65.6 7000.8
+ Ldl_Hdl   1     1.12 65.6 7000.8
+ Smoking   1     1.12 65.6 7000.8
+ Drinking  1     1.12 65.6 7000.8
+ Fasting   1     1.12 65.6 7000.8
+ Hypertension 1     1.12 65.6 7000.8
+ Stroke    1     1.12 65.6 7000.8
+ Cancer    1     1.12 65.6 7000.8
+ Bmi       1     1.12 65.6 7000.8

Step
```

```
No Standardized coefficients with Bonferroni p < 0.05
Response = Diabetes
  Estimate Standardized p-value Bonferroni p
(Int) -0.09553        0.00000  0.9000    1.0000
```

Bonferroni p-adjusted p.value < 0 significant, no worries

```
#[38;1;] # An influential observation
```

```
# A tibble: 1 x 10
  PerMatf.. Age Weight Height Weight.Fat Neck Chest Abdomen  Hip Thigh
      <dbl> <int>  <int>  <int>    <int> <int> <int>  <int> <int> <int>
1      21.3    50     80    170       30    34    94     90    99    59
# 1 more variable: Knee <dbl>, Ankle <dbl>, Biceps <dbl>, Forearm <dbl>,
  Wrist <dbl>
```

```
#[38;1;] # Outlier
```

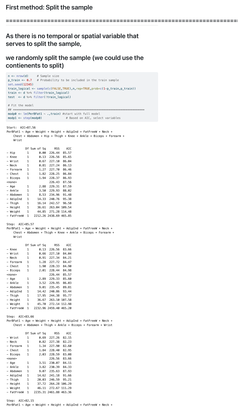
```
# A tibble: 1 x 10
  PerMatf.. Age Weight Height Weight.Fat Neck Chest Abdomen  Hip Thigh
      <dbl> <int>  <int>  <int>    <int> <int> <int>  <int> <int> <int>
1      35.8    22    206     64       45    51   148    132   74    42
# 1 more variable: Knee <dbl>, Ankle <dbl>, Biceps <dbl>, Forearm <dbl>,
  Wrist <dbl>
```

Interpretation

```
==========================================================================================
```

```
##> Graphical Inference
```

```
pairs(PerMatfat[num])
```




```
Just right curves
```

```
##> Marginal response curves see predict outcome, expected values + uncertainty
```

```
newsamp2 <- newsamples(f, Weight,seq(floor(Weight*0.8),ceiling(Weight*1.2)))
head(newsamp2)
```

```
Weight.mean   SD ymf.lower CI.upper CI.
  [1,]  22.2 0.8200 200   18.5  25.9
  [2,]  23.8 0.7800 200   19.1  25.9
  [3,]  25.4 0.7400 200   19.7  25.9
  [4,]  27.0 0.7000 200   20.3  25.9
  [5,]  28.6 0.6600 200   20.9  25.9
  [6,]  30.2 0.6200 200   21.5  25.9

Confidence level used: 0.95
```

```
pairs(newsamp2) #check all exact predictions pairwise
```

```
contrast          estimate  SE df t.ratio p.value
Weight0.8 - Weight0.9  -1.6 0.4 197  -3.94  0.0001
Weight0.8 - Weight1.0  -3.2 0.8 197  -3.94  0.0001
Weight0.8 - Weight1.1  -4.8 1.2 197  -3.94  0.0001
Weight0.8 - Weight1.2  -6.4 1.6 197  -3.94  0.0001
Weight0.9 - Weight1.0  -1.6 0.4 197  -3.94  0.0001
Weight0.9 - Weight1.1  -3.2 0.8 197  -3.94  0.0001
Weight0.9 - Weight1.2  -4.8 1.2 197  -3.94  0.0001
Weight1.0 - Weight1.1  -1.6 0.4 197  -3.94  0.0001
Weight1.0 - Weight1.2  -3.2 0.8 197  -3.94  0.0001
Weight1.1 - Weight1.2  -1.6 0.4 197  -3.94  0.0001
Weight0.8 - Weight0.9  -1.6 0.4 197  -3.94  0.0001
Weight0.8 - Weight1.0  -3.2 0.8 197  -3.94  0.0001
Weight0.8 - Weight1.1  -4.8 1.2 197  -3.94  0.0001
Weight0.8 - Weight1.2  -6.4 1.6 197  -3.94  0.0001
Weight0.9 - Weight1.0  -1.6 0.4 197  -3.94  0.0001
Weight0.9 - Weight1.1  -3.2 0.8 197  -3.94  0.0001
Weight0.9 - Weight1.2  -4.8 1.2 197  -3.94  0.0001
Weight1.0 - Weight1.1  -1.6 0.4 197  -3.94  0.0001
Weight1.0 - Weight1.2  -3.2 0.8 197  -3.94  0.0001
Weight1.1 - Weight1.2  -1.6 0.4 197  -3.94  0.0001

P value adjustment: tukey method for comparing a family of 6 estimates
```

```
# # Good Prediction !!! use MC-check #
#%%%
```